MA3521 — In-Class Worksheet Tuesday/Wednesday

Name:

Do not solve explicitly for y.

1. Linear $\quad ty' + 2y = e^t \quad y(1) = 0$

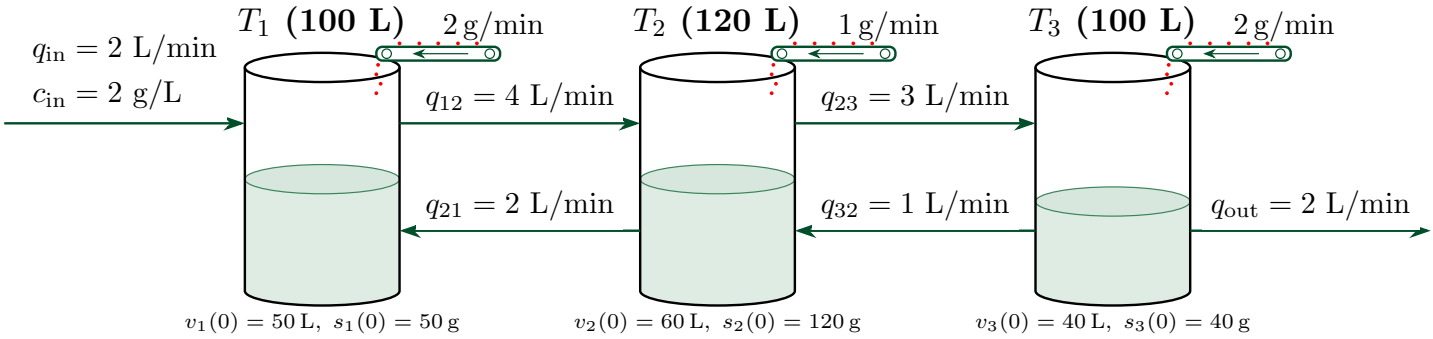
   Suitable intermediate step

   Answer

2. Separable $\quad y' = 2t\cos(t^2)(1 + y^2) \quad y(0) = 0$

   Suitable intermediate step

   Answer



Tanks are well-mixed. Time t is in min, volumes $\mathbf{v}(t)$ in L, and salt masses $\mathbf{s}(t)$ in g. Conveyors add dry salt at the rates shown.

Set up $\mathbf{s}' = \boldsymbol{A}\mathbf{s} + \mathbf{b}$. Do not row-reduce or solve.

1. $\mathbf{v}(t) =$ [ ] L $\quad \mathbf{s}(0) =$ [ ] g

2. $\boldsymbol{A} =$ [ ] min^{-1} $\quad \mathbf{b} =$ [ ] g/min

3. A suitable UC form is $\mathbf{s}_p(t) =$ [ ]

4. To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$ [ ]